LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Household Carbon Footprint & Target Tracker

NAME	**Aditya R Badde**
SRN	**PES2UG25CS802**
SEC	**A**
PROBLEM STATEMENT	Household Carbon Footprint & Target Tracker

Interface Description

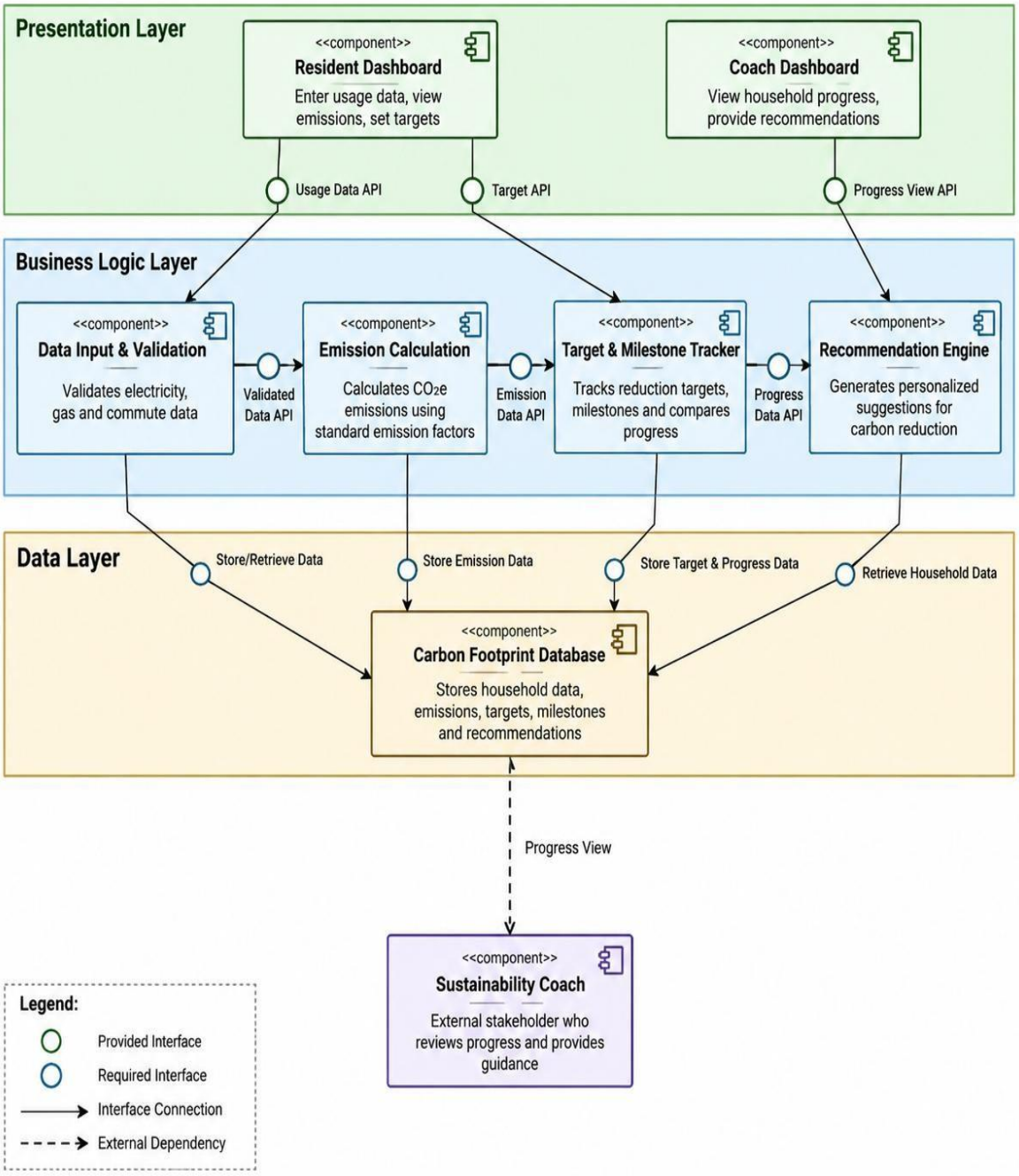
INTERFACE	INTERFACE PURPOSE
Activity Data API	Sends household electricity, gas, and commute data from the Carbon Footprint Dashboard to Data Input & V
Validated Activity API	Sends validated household activity data from Data Input & Validation to the Emission Calculator.
Emission Result API	Sends calculated CO2e emissions from the Emission Calculator to the Target & Milestone Tracker.
Target Progress API	Sends target progress and milestone status from the Target & Milestone Tracker to the Carbon Footprint Das
Carbon Data Query	Stores and retrieves activity data, calculated emissions, reduction targets, and milestone progress from the C
Coach Progress API	Provides household carbon-emission and target-progress information to the Sustainability Coach for review.

UML Component Diagram - Layered Architecture

HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER

UML Component Diagram - Layered Architecture

UML Component Diagram - Layered Architecture
HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER
UML Component Diagram - Layered Architecture



Component Identification

Carbon Footprint Dashboard: Presentation layer: allows residents to enter electricity, gas, and commute data and view calculated emissions, targets, and milestone progress.

Data Input & Validation: Business layer: validates household activity inputs such as electricity consumption, gas usage, and commute distance before processing.

Emission Calculator: Business layer: converts validated activity data into household CO2e emissions using the applicable emission factors.

Target & Milestone Tracker: Business layer: compares calculated emissions with reduction targets and determines milestone/target progress.

Carbon Footprint Database: Data layer: stores activity data, calculated emissions, reduction targets, and milestone progress for retrieval and tracking.

Sustainability Coach: External stakeholder: reviews household emissions and target progress and supports monitoring of carbon-reduction goals.

Architecture Selection

We chose Layered Architecture for the Household Carbon Footprint & Target Tracker.

Architectural Choice: The system uses Presentation, Business Logic, and Data layers. The Presentation layer provides the carbon footprint dashboard. The Business Logic layer handles activity-data validation, CO2e calculation, target comparison, and milestone tracking. The Data layer stores household activity, emission, target, and progress information.

Reason 1: Layered Architecture provides clear separation of concerns. User interaction is separated from data validation, carbon-emission calculation, target tracking, and data storage, making the system easier to understand, test, maintain, and modify.

Reason 2: The architecture allows important functions such as emission-factor calculations, target comparison, and milestone rules to be modified or improved within the Business Logic layer without changing the entire system.

Security Advantage: The Presentation layer does not directly access stored household data. Authentication and authorization for Residents and Sustainability Coaches can be enforced in the Business Logic layer, while household activity and emissions data in transit can be protected using TLS 1.2 or higher.

Performance Benefit: Emission calculation and target-progress evaluation are isolated in the Business Logic layer, allowing these operations to be optimized independently. This supports responsive calculation and progress tracking as household activity records increase.

Conclusion: Layered Architecture provides a clear structure for the system by separating user interaction, carbon-emission and target processing, and persistent data. It supports maintainability, security, and efficient processing while keeping the system organized and easy to extend.